

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

MULTIPLE INTEGRALS

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0.1 The Riemann integral of a function of a single variable

Before we start the presentation on multiple integrals, it is appropriate to recall the steps for the construction of the Riemann integral of a function of a single variable over a closed bounded interval. As a matter of illustration, we consider a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$. We want to define the (definite) Riemann integral of the function f over the closed bounded interval $[a, b]$.

This is a step by step process:

- (1) We consider a “partition” $P = \{x_0 = a, x_1, \dots, x_n = b\}$ of the interval $[a, b]$ into n -subinterval. That is,

$$P : x_0 = a < x_1 < \dots < x_{n-1} < x_n = b.$$

The subinterval $[x_{i-1}, x_i]$ has length $\Delta x_i = x_i - x_{i-1}$.

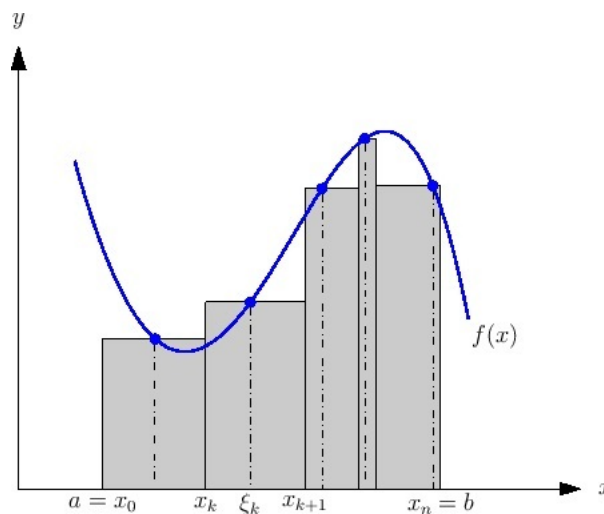


Figure 1: The Riemann integral of the continuous function of a single variable $f : [a, b] \rightarrow \mathbb{R}$.

- (2) We choose a sample number ξ_i in each subinterval $[x_{i-1}, x_i]$ and we write down the sum

$$\sum_{i=1}^n f(\xi_i) \Delta x_i = \sum_{i=1}^n f(\xi_i) (x_i - x_{i-1})$$

called a **Riemann sum corresponding to the partition P and to the choice of the sample number ξ_i in the sub-interval $[x_{i-1}, x_i]$.**

- (3) When we consider a new partition P' containing the set of number x_i from the previous partition, we obtain a **refinement** of the partition P . Notice that the maximum length of the subintervals in P' is smaller than the maximum length in P . This is due to the fact that each subinterval in the refined partition P' is contained in some subinterval in the original partition P . Now, if we continue this process in which at each step we refine the previous partition, we obtain that the corresponding sequence of Riemann sum converges to a certain number which is called the **Riemann integral of the function f over the interval $[a, b]$.**

Notice, that for continuous functions such a limit always exists and is independent of the partition and of the choice of the sample number ξ_i .

In general, such a limit might not exist for discontinuous functions. The existence of the limit is called the **Riemann integrability of the function f over the interval $[a, b]$.** This concept will be rigourously revisited in the second semester module Real Analysis (2RA).

Now, the goal to define the Riemann double and triple integrals (for functions of several variables) is to mimic the process described above.

0.2 The construction of the Riemann double integral

0.2.1 The area of a domain in $\mathbb{R}^2$

Before we develop the concept of the Riemann integral for functions of two variables, it will be helpful to get some feeling for a notion of area of a domain in $\mathbb{R}^2$.

If the domain is one of the usual geometric figures like rectangles, circles and other classical polygonal domains, then we will assume that its area is known from “elementary” classical geometry.

Now, let D be any bounded domain in $\mathbb{R}^2$. Let $R = [a, b] \times [c, d]$ be a rectangle containing the domain D . Let us partition the intervals $[a, b]$ and $[c, d]$ as:

$$\begin{cases} a = x_0 < x_1 < \dots < x_{n-1} < x_n = b, \\ c = y_0 < y_1 < \dots < y_{m-1} < y_m = d. \end{cases}$$

By drawing the vertical lines $x = x_i$ ($i = 0, \dots, n$) and the horizontal lines $y = y_j$ ($j = 0, \dots, m$) we obtain a so-called grid G over the domain D for which the diameter of the largest subrectangle is called a **mesh** and is denoted by $\text{mesh}(G)$. We can split this family of subrectangles in the grid G into three classes:

- (i) the subrectangles with no points in common with D ;
- (ii) the subrectangles containing only the interior points of D (no boundary point);
- (iii) the subrectangles containing at least one boundary point of D .

Now the strategy to find an approximation of the area $A(D)$ of the domain D is to ignore those subrectangles in the class (i) and to introduce the numbers:

$$\mathcal{I}(D, G) := \text{Sum of the areas of the subrectangles in class (ii)} \quad (1)$$

called the **inner area** of D relative to G ;

$$\mathcal{O}(D, G) := \mathcal{I}(D, G) + \text{Sum of the areas of the subrectangles in class (iii)} \quad (2)$$

called the **outer area** of D relative to G ;

The natural question which arises is to ask for the grid G which realises the best approximation of the area of the domain D . This question is reformulated as

$$\boxed{\text{finding the grid } G \text{ for which the number } \mathcal{O}(D, G) - \mathcal{I}(D, G) \text{ is as smaller as possible}}. \quad (3)$$

Notice that

$$0 \leq \mathcal{I}(D, G) \leq \mathcal{O}(D, G) \leq (b-a)(d-c).$$

Similarly to what has been mentioned above for the integral of a function of a single variable, we will realize that we obtain a better approximation when we refine the grid G because the mesh of the refinement is smaller than the one of G . In fact, each subrectangle in the refined grid is contained in some subrectangle in the original grid.

0.2.2 The double integral

The double integral in a rectangular domain:

Let $R = [a, b] \times [c, d]$ and let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function (as in the case of the Riemann integral of a function of a single variable).

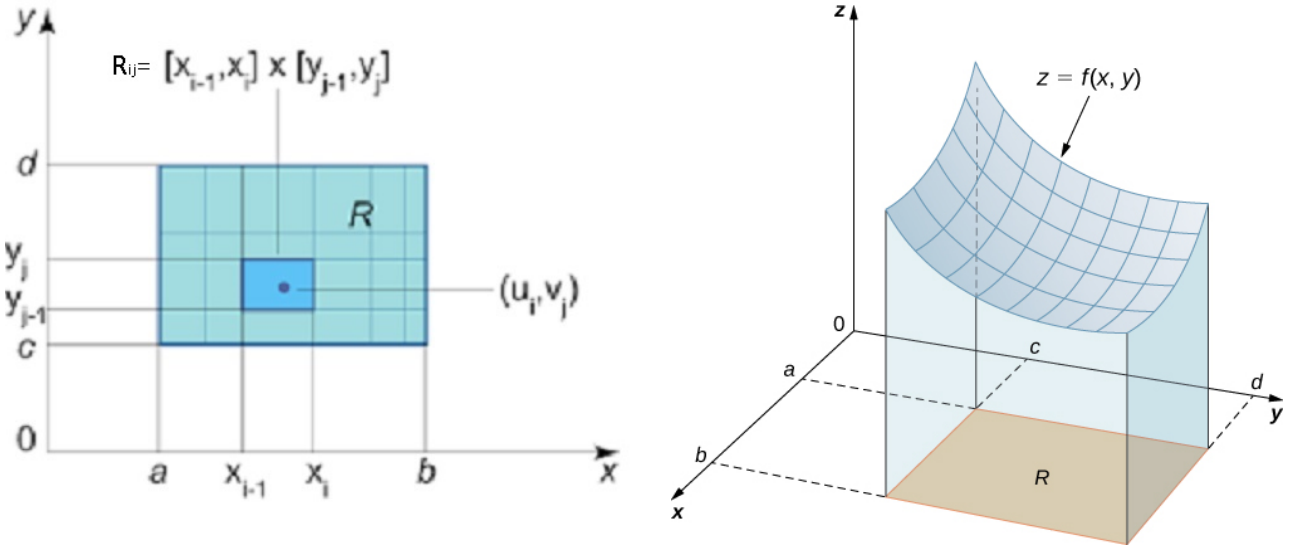


Figure 2: **Left:** A partition of the rectangle $R = [a, b] \times [c, d]$ into mn subrectangle $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ with the choice of sample points $(u_i, v_j) \in R_{ij}$. **Right:** The graph of the continuous function $f : R \rightarrow \mathbb{R}$.

- (1) We consider the grid G as before and obtain a “partition” of R into nm subrectangle $R_{ij} = [x_{i-1}, x_i] \times [y_{j-1}, y_j]$ with area $A(R_{ij}) = \Delta x_i \Delta y_j$. Now, on each subrectangle we choose a sample point $(u_i, v_j) \in R_{ij}$.
- (2) We write down the **Riemann sum corresponding to the grid G and the choice of the sample point $(u_i, v_j) \in R_{ij}$** as

$$\sum_{i=1}^n \sum_{j=1}^m f(u_i, v_j) A(R_{ij}) = \sum_{i=1}^n \sum_{j=1}^m f(u_i, v_j) \Delta x_i \Delta y_j. \quad (4)$$

- (3) We want to pass to the limit in (4) to get a number called the **Riemann double integral of f over the rectangle D** denoted by

$$\iint_R f(x, y) dx dy = \lim_{m, n \rightarrow \infty} \sum_{i=1}^n \sum_{j=1}^m f(u_i, v_j) \Delta x_i \Delta y_j. \quad (5)$$

Indeed, it follows from the continuity of the function f that the limit above exists and is independent of the grid and of the choice of the sample point (u_i, v_j) . In general, the limit in (5) does not always exist for discontinuous functions. Whenever such a limit exists for some function f , we say that f is **Riemann integrable over R** through the concepts of **Riemann upper and lower sums** and also the concepts of **Riemann upper and lower integrals** well-known in Real Analysis. The concept of Riemann integrability for multiple integrals is also beyond the scope of these lecture notes.

Our interest in this course is on the evaluation of **double** and **triple integrals**. The evaluation of a double integral as we will see later, depends on the domain over which the function is integrated as is based on the process of successive integration of functions of a single variable. That is, in order to evaluate a double, we will first rewrite that double integral as repeated integrals for which the choice of the limits depends on the domain D . In the present case of a rectangular domain D , the choice of the limits is trivial. In fact,

$$\iint_R f(x, y) dx dy = \int_c^d \left(\int_a^b f(x, y) dx \right) dy = \int_a^b \left(\int_c^d f(x, y) dy \right) dx. \quad (6)$$

The equations in (6) are given within what Stewart calls the **first Fubini's theorem**.

Example 0.1 Evaluate the double integral $\iint_R (xy^2 - y) dx dy$ where $R = [0, 1] \times [1, 2]$.

Solution: Following (6) we get

$$\begin{aligned} \iint_R (xy^2 - y) dx dy &= \int_0^1 \left(\int_1^2 (xy^2 - y) dy \right) dx = \int_0^1 \left[\frac{1}{3} xy^3 - \frac{1}{2} y^2 \right]_1^2 dx \\ &= \int_0^1 \left[\left(\frac{8}{3} x - 2 \right) - \left(\frac{1}{3} x - \frac{1}{2} \right) \right] dx = \int_0^1 \left[\frac{7}{3} x - \frac{3}{2} \right] dx \\ &= \left[\frac{7}{6} x^2 - \frac{3}{2} x \right]_0^1 = \frac{7}{6} - \frac{3}{2} = -\frac{1}{3}. \end{aligned}$$

The next set of notes will be on double integrals on non-rectangular domains!